



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2019

Bachelor in Computer Applications
Course Title: System Analysis & Design
Code No: CACS 203
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Centre:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10×1 = 10]

2. Circle (O) the correct answer.

- i) Which of the following Information systems are aimed at improving the routine business activities on which all organizations depend?
 - a) Management Information systems
 - b) Decision support systems
 - c) Transaction Processing Systems
 - d) Executive Information
- ii) The project life cycle consists of
 - a) Understanding the scope of the project
 - b) Objectives of the project
 - c) Formulation and planning various activities
 - d) Word Processor
- iii) Which is the most important feature of spiral model?
 - a) Quality Management
 - b) Efficiency Management
 - c) Risk Management
 - d) Performance Management
- iv) includes the existing system, the proposed system, system flow charts, modular design of the system, print layout charts and data file designs.
 - a) Feasibility Report
 - b) Functional Specification Report
 - c) Design Specification Report
 - d) Terms of Reference
- v) For the best Software model suitable for the project, in which of the phase the developers decide a roadmap for project plan?
 - a) Software
 - b) System Analysis

c) Coding

d) Testing

vi) Using the approach, a new system is tested in one part of the organization before being implemented in others.

a) Direct

b) Parallel

c) phased

d) pilot

vii) extends the software beyond its original functional requirements.

a) Adaptive maintenance

b) Perfective maintenance

c) Corrective maintenance

d) Preventive maintenance

viii) Which normal form looks at removing partial dependencies?

a) First Normal Form

b) Second Normal Form

c) Third Normal Form

d) Fourth Normal Form

ix) In constructing ER diagrams, double ovals are used to denote

a) Multi-value table

b) Multi-value entity

c) Multi-value attributes

d) Multi-value key

x) Testing beyond normal operational capacity is

a) Performance testing

b) Stress testing

c) Recovery testing

d) None of the above



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Group B

Attempt any SIX questions.

[6×5 = 30]

3. When would you use agile methodologies? How is it different from waterfall approach to system development?
4. Why is project management important? Briefly explain the activities performed by the project manager during project execution.
5. List various methods of interacting with a system. Briefly explain the factors to be considered while designing a form?
6. What are the deliverables from coding and testing? Briefly explain the different approaches to installation.
7. Why is normalization required? State second normal form and explain it with a proper example.
8. Construct an E-R Diagram for football club that has a name and a ground and is made up for players. A player can play for only one club and a manager identified by his name manage a club. A footballer has a registration number, name and age. A club manager also buys players. Each club plays against other clubs in the league and matches have a date, venue and score.
9. Maintenance is an on-going process. Do you agree? Explain the process of maintaining information systems.

Group C

Attempt any TWO questions.

[2×10 = 20]

10. Develop a context diagram and top level logical DFD for the system made up of the following.

B & B is a mail-order company that distributes CDs, DVDs of music, games, movies, software at discount prices to club members.

- When an order processing clerk receives an order form, he or she verifies that the sender is a club member by checking the Member file.
- If the sender is not a member, the clerk returns the order along with a membership application form.
- If the customer is a member, the clerk verifies the order item data by checking the item file.
- Then the clerk enters the order data and saves it to the Daily Order file. The clerk also prints an invoice and shipping list for each order, which are forwarded to Order Fulfillment Department.

10. With proper reasoning, explain how CASE Tools aid in information system development? You have been hired as a system analyst in TU tech software development company and you are asked to analyze the way system works. What qualities do you need to have to analyze such type of systems?

11. a) Why software project often fails? Explain different types of software testing?
- b) List of OOAD. Differentiate between structured methodologies and object oriented methodologies.